

COS 214 Practical 6

TIVIDY: Workflow Management System

TASK 1: Research Workflow Management Systems

A workflow management system (WFMS) is a software system that manages the flow of work between users according to formal specifications of business processes called workflows. A workflow specifies what tasks need to be performed and decides in what order they need to be executed in. Furthermore, it decides how data flows internally between the tasks and the applications needed for the tasks to be carried out. An example is a travel expense reimbursement workflow will specify the tasks of filling, checking and signing the form and reimbursing the amount of money needed to be reimbursed. The workflow specifies that before the amount is reimbursed the form first needs to be signed. It also defines the flow of the expense form between the tasks and the spreadsheet application. In addition to that for each task of the workflow, a rule needs to be set that specifies the responsible and capable users are able to carry out the task.

A workflow definition is the static specification of the process or the template of a process, activity or an automatic part of a manual activity. It is built from available building blocks and once the processes and activities have been defined, the workflow system can assist in the execution of these processes. However, a running workflow instance is for a particular execution of a process. A workflow definition would be an object class and the workflow instance would be the instance of that class.

Activities or work items are modelled and represented at two different levels. At definition level an activity describes a logical step in the process-its name, required role, preconditions and routing. It is a reusable description shared by many instances. At runtime, each activity in a running instance is represented by a work item, which combines the activity with a specific case. The work item has its own state, assigned participant, data and history.

Work items could be assigned and routed by evenly distributing work among all available resources or give the waiting work to a resource with the shortest queue or manually implement assignments, however this is not optimized. The WFMS needs to know what is available to perform the work and have a profile about each user which includes what each resource is qualified to do and how good it is at that type of work and whether the supervisor wants the work to be assigned to them. As for routing there are two types of routing which is control flow routing and resource routing, Control flow routing defines the path through the process which could be sequential, parallel or conditional. Resource routing deals with which participant receives the work item which could be role based, workload based and rule based. More advanced systems use dynamic routing which results in escalation when deadlines are missed and delegation to another qualified member. A well designed workflow engine separates these two so they can be changed independently.

In a workflow management system approvals can be handled by routing a work item to one or more authorised participants for a formal decision. It could be one approval participant or sent through many participants for approval or multiple participants could decide simultaneously. Rejections could terminate or loop back for reprocessing or trigger a response. Lastly in the case of Escalation it could be triggered by a time limit which uses hierarchical routing to route the work item to a supervisor participant which may have a maximum number of escalation levels.

Roles are represented by assigning a role or roles needed for a step. Roles are abstract job functions. Participants are the real people or systems assigned to those roles at runtime and the responsibilities are the tasks the role/s or participant/s are expected to do.

Dependencies are handled by defining the order in which work items must happen which are implemented by conditional logic. Events are handled by having event listeners and real time observability. Failures are handled by rollbacks, retrying or aborting the process.

Workflow management systems usually communicate to external systems using APIs and adapters because it needs to get something it doesn't have on its own.

Workflow history and audit information are recorded by capturing meaningful action, state change and decisions that happen during a workflow instance and storing them in a timestamped log which is usually stored in an append-only structure.

Running workflows are managed by watching several live things such as the status of each instance, progress, bottlenecks, errors and failures and time metrics. Reporting usually happens with dashboards that summarize what has happened over time and notifying observing participants of any changes.

A workflow engine is reusable across different processes if it only contains generic logic and if it pushes all process specific details into data or configuration. The engine should know how to run a workflow but not know what the workflow is. It should have configurable variation points.

The workflow definition should have information such as the process structure, order and routing, roles, rules, data schema and transitions. The workflow instance contains the actual data, state and history for one specific case.

Real workflow systems model nested activities and subb-processes by embedding sub processes in a tree structure or by call activity where the parent process calls a separate process.

Configurable decisions in a workflow system are assignment, routing, approval, escalation, deadlines, notifications, conditions and integration.

Changes to workflow definition while work is already in process is handled by versioning with instance pinning. The running instance never changes its definition. It keeps using the version it started with and a new version is created for the new definition.

References:

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